PROGRAM 6 – VACUUM CLEANER PROBLEM

AIM

To write a Python program to implement the Vacuum Cleaner problem using an intelligent agent.

ALGORITHM

1. Start the program.
2. Define two rooms, A and B.
3. Represent each room as Clean or Dirty.
4. Read the initial condition of both rooms.
5. Place the vacuum cleaner in room A.
6. Check whether the current room is dirty.
7. If dirty, clean the room.
8. Move the vacuum cleaner to the next room.
9. Repeat the cleaning process.
10. Display the final condition of both rooms.

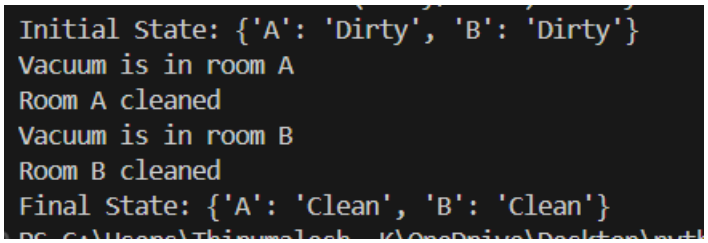
CODE

```
rooms={"A":"Dirty","B":"Dirty"}
location=input("Enter vacuum location (A/B): ").upper()
rooms["A"]=input("Enter status of room A (Dirty/Clean): ").capitalize()
rooms["B"]=input("Enter status of room B (Dirty/Clean): ").capitalize()
print("Initial State:",rooms)
for room in ["A","B"]:
    location=room
    print("Vacuum is in room",location)
    if rooms[location]=="Dirty":
        rooms[location]="Clean"
        print("Room",location,"cleaned")
    else:
        print("Room",location,"is already clean")
print("Final State:",rooms)
```

INPUT

```
Enter vacuum location (A/B): A
Enter status of room A (Dirty/Clean): Dirty
Enter status of room B (Dirty/Clean): Dirty
```

OUTPUT



RESULT

Thus, the Vacuum Cleaner problem was successfully implemented using a simple intelligent agent in Python.